

Chi Square Test

- Used in inferential statistics.

The chi square test for goodness of Fit test claims about population proportion { categorical variables } $\rightarrow$ Ordinal, normal.

In 2010 census of the city, the wt of the individuals in a small city were found to be the following

< 50 kg	50-75	> 75
20%.	30%.	50%.

In 2020, ages of $n = 500$ individuals were sampled. Below are the results.

< 50	$50 - 70$	> 75
140	160	200

Using $\alpha = 0.05$, would you conclude the population difference of weights has changed in the last 10 years

Ans

2010

Expected

< 50	$50 - 75$	> 75
20%	30%	50%

2020, $n = 500$

observation

< 50	$50 - 75$	> 75
140	160	200

Expected

< 50	$50 - 75$	> 75
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$0.2 \times n$	$0.3 \times n$	$0.5 \times n$
$= 100$	$= 150$	$= 250$

But confidence interval is given as 95% so we are not concluding directly. We will use chi square test

① Null Hypothesis: H_0 = The data meets the expectations

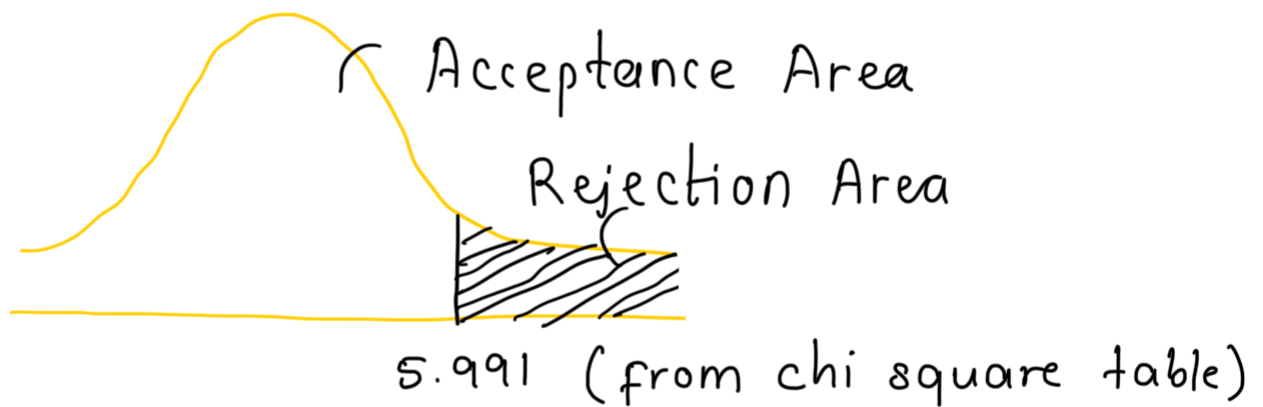
Alternate Hypothesis: H_1 = The data does not meet expectations

② $\alpha = 0.05 \therefore CI = 95\%$

③ Degree of Freedom

$$df = k - 1 = 3 - 1 = 2$$

④ Decision Boundary



5) IF K^2 is greater than 5.99, Reject H_0
else we Fail to reject the null hypothesis.

$$\begin{aligned} \chi^2 &= \frac{\sum (O - E)^2}{E} \\ &= \frac{(140 - 100)^2}{100} + \frac{(160 - 150)^2}{150} + \frac{(200 - 250)^2}{250} \\ &= \frac{40 \times 40}{100} + \frac{10 \times 10}{150} + \frac{50 \times 50}{250} \\ &= 16 + 2 + 10 \end{aligned}$$

$$= 26 + \frac{2}{3}$$

$$= \frac{26 \times 3 + 2}{3} = \frac{78 + 2}{3} = \frac{80}{3}$$

$$\underline{\underline{\chi^2 = 26.66}} > 5.99$$

$\therefore$ We reject the H_0

Conclusion: There is a population difference & it has increased when compared to census in 2010

Table 1: Choosing a Chi-square test

	<u>Chi-Square Goodness of Fit Test</u>	<u>Chi-Square Test of Independence</u>
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Number of variables	One	Two
Purpose of test	Decide if one variable is likely to come from a given distribution or not	Decide if two variables might be related or not
Example	Decide if bags of candy have the same number of pieces of each flavor or not	Decide if movie goers' decision to buy snacks is related to the type of movie they plan to watch
Hypotheses in example	Ho: proportion of flavors of candy are the same Ha: proportions of flavors are not the same	Ho: proportion of people who buy snacks is independent of the movie type Ha: proportion of people who buy snacks is different for different types of movies
Theoretical distribution used in test	Chi-Square	Chi-Square
Degrees of freedom	Number of categories minus 1	Number of categories for first variable minus

In our example, number of flavors of candy minus 1	1, multiplied by number of categories for second variable minus 1 In our example, number of movie categories minus 1, multiplied by 1 (because snack purchase is a Yes/No variable and $2-1 = 1$)
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